

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS (NC State, May 2026). Built **compliance-ready RAG for healthcare/fintech** (per-tenant data isolation, deterministic outputs) and production LLM pipelines across **3 providers and 8+ models** (Claude, GPT-4o, Gemini) at an AI startup. Full-stack ownership from DB schema to UI -- TypeScript, Node.js, React, Python, PostgreSQL, Redis, Temporal.io. Healthcare claim automation at Mphasis. Ready to ship AI-powered clinical review features at Brellium.

WORK EXPERIENCE

Isomer AI May 2025 -- Present

Product Engineer Intern

- Owned full backend of a production AI platform -- built **multi-model LLM pipelines across 3 providers and 8+ models** (Claude, GPT-4o, Gemini) via MCP with Temporal.io durable workflow orchestration; provider-agnostic interfaces so any model is swappable without downstream changes.
- Shipped 10+ features end-to-end using **TypeScript, Node.js, Express, Prisma ORM, PostgreSQL, Redis**, Auth0, Docker, Sentry; document processing pipelines (PDF gen, email parsing, S3); real-time event pipelines and PR reviews across the team.

Mphasis (HP Company) Jun -- Aug 2024

Software Developer Intern

- Built **health claim automation** workflow (OCR, NLP, ML) improving accuracy 25%; automated deployments on AWS and GCP with Grafana + Sentry observability and red-teaming support.

Capgemini Jun -- Aug 2023

AI/ML Intern

- Kafka + Spark + Hadoop pipelines for real-time data ingestion; cut processing time 15% and latency 20%.

PROJECTS

Glance AI -- Privacy-First RAG System 2025

Compliance-ready RAG for fintech and healthcare: per-tenant data isolation at the vector retrieval layer prevents cross-tenant context leakage and LLM hallucinations. Deterministic output architecture ensures factual correctness for regulated data environments -- same problem Brellium solves for clinical compliance. Python, LLMs, Qdrant, Vector Databases, RAG

LLM Eval Harness 2025

Vendor-neutral benchmark comparing Claude, GPT-4o, and Gemini on 2,000-example frozen test set. Tracked **grounded hallucination rate, p50/p95 latency, cost/call**, and abstention accuracy with bootstrap significance testing -- critical for AI in regulated clinical settings. Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker

AlgoPulse -- Agentic AI Platform 2025

Event-driven LLM orchestration platform. Diagnosed API-polling bottleneck; pivoted to Supabase Edge Functions streaming AI reasoning to client. **Temporal.io** for durable workflow state + Kafka for fault-tolerant coordination -- **2,000 async req/sec** without state loss. TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

EDUCATION

MS Computer Science -- North Carolina State University Aug 2024 -- May 2026

AI, Operating Systems, Data Science, Algorithms, Software Engineering

B.Tech Computer Science -- Mumbai University Aug 2020 -- May 2024

CGPA 9.32/10 | Machine Learning, Big Data Infra, Data Warehousing, Algorithms

SKILLS

Languages: Python, TypeScript, Java, JavaScript, SQL, C

AI/ML: OpenAI API, Claude API, Gemini API, LangChain, Hugging Face, PyTorch, RAG, Qdrant, MCP, LLM Evals

Stack: Node.js, React, Express, Prisma ORM, Temporal.io, Kafka, FastAPI, Next.js

Infra: Docker, AWS, GCP, PostgreSQL, Redis, MongoDB, Sentry, Prometheus, GitHub Actions

Awards: HackNC24 Top 10 (100+ teams), 6+ national hackathon wins, IEEE Published, Top 1% Sanjeevani (1100+ teams)